Clinical Preprocessing Plan After Splitting

Zero Point

We already have:

`Train`
`Validation`
`Test`

All preprocessing methods that learn from the data must be **fitted on the training set only**, then applied to train, validation, and test.

1. Rule-Based Data Cleaning

Column: `[Performance Status]`

Convert invalid `Performance Status` values to missing values `(NaN)` in:

`Train`
`Validation`
`Test`

Allowed values are only: **0, 1, 2, 3, 4**

Any other value, such as: **80, 90, 100, ...** must be converted to: **NaN**

2. Missing Indicator Method

Columns:

`[Tobacco Consumption]`
`[Alcohol Consumption]`
`[Performance Status]`

Create a missing indicator column for each one:

`[Tobacco Consumption_missing]`
`[Alcohol Consumption_missing]`
`[Performance Status_missing]`

Each indicator means:

```
1 = the original value was missing
0 = the original value was present
```

This must be done before imputation.

We use missing indicators because these variables have important missingness patterns. This preserves the information that the original value was missing before imputation.

3. Median Imputation

Column: `[Age]`

Missing values in `Age` are replaced by the median age calculated from the **training set only**.

Then the same training median is applied to:

```
Train
Validation
Test
```

We use median imputation for `[Age]` because it is a continuous numerical variable.

4. Most-Frequent Imputation

Columns:

```
[Gender]
[Tobacco Consumption]
[Alcohol Consumption]
[Treatment]
[Performance Status]
```

Missing values are replaced by the most frequent value, also called the mode, calculated from the training set only.

Then the same training-based value is applied to:

```
Train
Validation
Test
```

We use most-frequent imputation for binary and ordinal/discrete clinical variables because these variables are categorical or discrete, not continuous.

5. Constant Imputation

Column: `[M-stage]`

Missing values in `[M-stage]` are replaced with a fixed category:

```
Unknown
```

Example:

```
[M-stage] = NaN -> [M-stage] = Unknown
```

We use constant imputation for `[M-stage]` because it is a categorical variable.

6. Standardization

Column: `[Age]`

After median imputation, standardize `[Age]` using z-score standardization:

```
Age_scaled = (Age - mean_train_age) / std_train_age
```

The mean and standard deviation must be calculated from the **training set only**.

Then apply the same scaler to:

```
Train
Validation
Test
```

Only `[Age]` should be standardized. Do not standardize binary variables or missing indicators.

7. One-Hot Encoding

Column: `[M-stage]`

After replacing missing values with `"Unknown"`, apply one-hot encoding.

Example output columns:

```
M-stage_M0
M-stage_M1
M-stage_Mx
M-stage_Unknown
```

The encoder must be fitted on the **training set only**, then applied to validation and test.

Use:

```
handle_unknown="ignore"
```

so validation/test do not crash if they contain a category not seen in training.